TCS349: Responsible and Explainable AI

Unit 2, Lecture 6

AI in Surveillance: Facial Recognition, Video & Behavioural Monitoring

How AI is used to watch, recognise and analyse people at scale - and the case examples that show both its capability and its risk.

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Facial Recognition

Video Analytics

Behavioural Monitoring

Ethics & Regulation

1. What Is AI-Enabled Surveillance?

AI-enabled surveillance uses machine learning to observe, identify or analyse people's behaviour, often continuously and at scale.

It spans a range of techniques: recognising faces, tracking movement, and inferring behaviour or intent from video and sensor data.

Surveillance AI is used by governments, law enforcement, employers and private businesses, each with different goals and oversight.

The same underlying techniques can support public safety or enable intrusive, unaccountable monitoring, depending on how they are governed.

2. Facial Recognition - How It Works

Facial recognition detects a face in an image or video, then converts its features into a numerical representation.

This representation is compared against a database of known faces to find a match or confirm an identity.

Systems operate in two modes: verification, matching one face to one claimed identity, and identification, searching a face against many.

Accuracy depends heavily on image quality, lighting, angle and the diversity of the data used to train the model.

3. Facial Recognition - Common Applications

Access control: unlocking phones or verifying identity at secure facilities.

Law enforcement: identifying persons of interest from CCTV or investigative footage.

Border and travel security: matching travellers to passport or visa photographs.

Retail and venues: identifying repeat visitors or flagged individuals at entry points.

4. Facial Recognition - Accuracy & Bias Concerns

Facial recognition systems have shown measurably lower accuracy for women and for people with darker skin tones in multiple independent studies.

This connects directly to the data and algorithmic bias concepts studied in Unit I - underrepresentation in training data produces unequal error rates.

A false match in identification mode can lead to wrongful suspicion or arrest, making accuracy gaps a serious real-world harm.

Subgroup-wise accuracy reporting, as introduced in Lecture 6, is now a standard expectation for facial recognition vendors.

5. Case Example: Facial Recognition in Law Enforcement

Several police departments have used facial recognition to match CCTV images against criminal databases during investigations.

Documented wrongful-arrest cases have involved individuals misidentified by the system, disproportionately affecting Black men in the reported cases.

These cases led some cities to restrict or ban police use of facial recognition, while others introduced mandatory human review of any match.

The case illustrates why accountability and human oversight, covered in Lectures 13-14, matter most in exactly these high-stakes deployments.

6. Video Analytics - What It Is

Video analytics applies AI to video streams to automatically detect objects, events or patterns, without a human watching every frame.

It goes beyond simple recording, turning raw footage into structured, searchable information such as counts, alerts and movement paths.

Common techniques include object detection, motion tracking and anomaly detection.

Unlike facial recognition, video analytics does not always require identifying a specific individual.

7. Video Analytics - Common Applications

Traffic management: counting vehicles, detecting congestion and identifying violations automatically.

Public safety: detecting unattended objects, crowd density or unusual movement patterns in public spaces.

Retail operations: analysing footfall, queue lengths and store layout effectiveness.

Industrial safety: detecting when a worker enters a hazardous zone without protective equipment.

8. Case Example: Video Analytics in Retail

A retail chain deployed video analytics to track customer movement patterns and optimise store layout and staffing.

The system flagged unusually long dwell times near high-value items, which staff used to prioritise customer assistance, not automatic suspicion.

Customers were not individually identified - only aggregated movement patterns were analysed and stored.

This design choice - analysing patterns rather than identities - meaningfully reduces the privacy risk compared to identity-based surveillance.

9. Behavioural Monitoring - What It Is

Behavioural monitoring uses AI to observe patterns in how people act over time, rather than identifying who they are in a single moment.

It can track physical actions, such as posture and movement, or digital actions, such as keystrokes, app usage or browsing patterns.

The goal is typically to infer a state or trend - attention, fatigue, productivity or risk - from repeated observation.

Behavioural monitoring often raises deeper concerns than one-time identification, since it can reveal habits people did not choose to disclose.

10. Behavioural Monitoring - Common Applications

Workplace productivity tools: tracking application usage, active time and work patterns on company devices.

Driver monitoring systems: detecting drowsiness or distraction using in-cabin cameras and sensors.

Online platforms: inferring engagement, mood or intent from clicks, scrolling and time spent on content.

Educational proctoring: monitoring student behaviour during online exams to flag potential dishonesty.

11. Case Example: Behavioural Monitoring in the Workplace

Some companies have deployed software that tracks employee keystrokes, active windows and idle time throughout the workday.

Employees in several reported cases were unaware of the extent of monitoring until it was disclosed internally or through investigative reporting.

The resulting concern was less about the technology itself and more about the lack of transparency and consent around its use.

This case connects directly to the transparency principle from Lecture 11: monitoring without disclosure undermines trust even when the goal is legitimate.

12. Ethical & Societal Concerns Across Surveillance AI

Chilling effect: awareness of being watched can change how people behave in public and private spaces, even without any wrongdoing.

Function creep: a system deployed for one narrow purpose is often expanded to new, unapproved uses over time.

Disproportionate impact: surveillance is frequently deployed more heavily in already-marginalised communities.

Consent gaps: many surveillance systems operate with limited or no meaningful consent from the people being observed.

13. Regulation & Responsible Deployment

Several cities and countries have introduced specific restrictions or bans on facial recognition in public spaces and policing.

Responsible deployment typically requires a documented purpose, a defined retention period for data, and independent oversight.

The accountability frameworks and human oversight levels from Lectures 13-14 apply directly to surveillance system design.

Lecture 16 continues this thread by examining the privacy principles that govern how observed data can be collected and used.